Google DeepMind AI Hackathon · Track 4 · Developer tools

Spec Gap Replay

Walk a spec’s documented paths.
Stop where the document can’t decide.

● Defined ■ Blocked by document ◆ Needs decision ○ Out of scope

The problem

A developer gets a PRD.
The happy path is clear. The edge cases aren’t.

> Can the person who started this approve it?

> What does this button change in this state?

> Two sections say different things. Which one wins?

Without a way to check, you guess, or you ask a vague question.

How it works

The model reads. Code decides.

Document
PRD, spec, policy

→

gemini-4-argon
rules + exact quotes

→

Code checks
quotes exist · references resolve

↺ failed structural checks return to Argon with the validation errors · up to 3 extraction attempts · checks don't prove meaning; a person reviews

A person
includes / excludes rules

→

Deterministic engine
applies the reviewed rules

→

gemini-3.8-flash
drafts the question

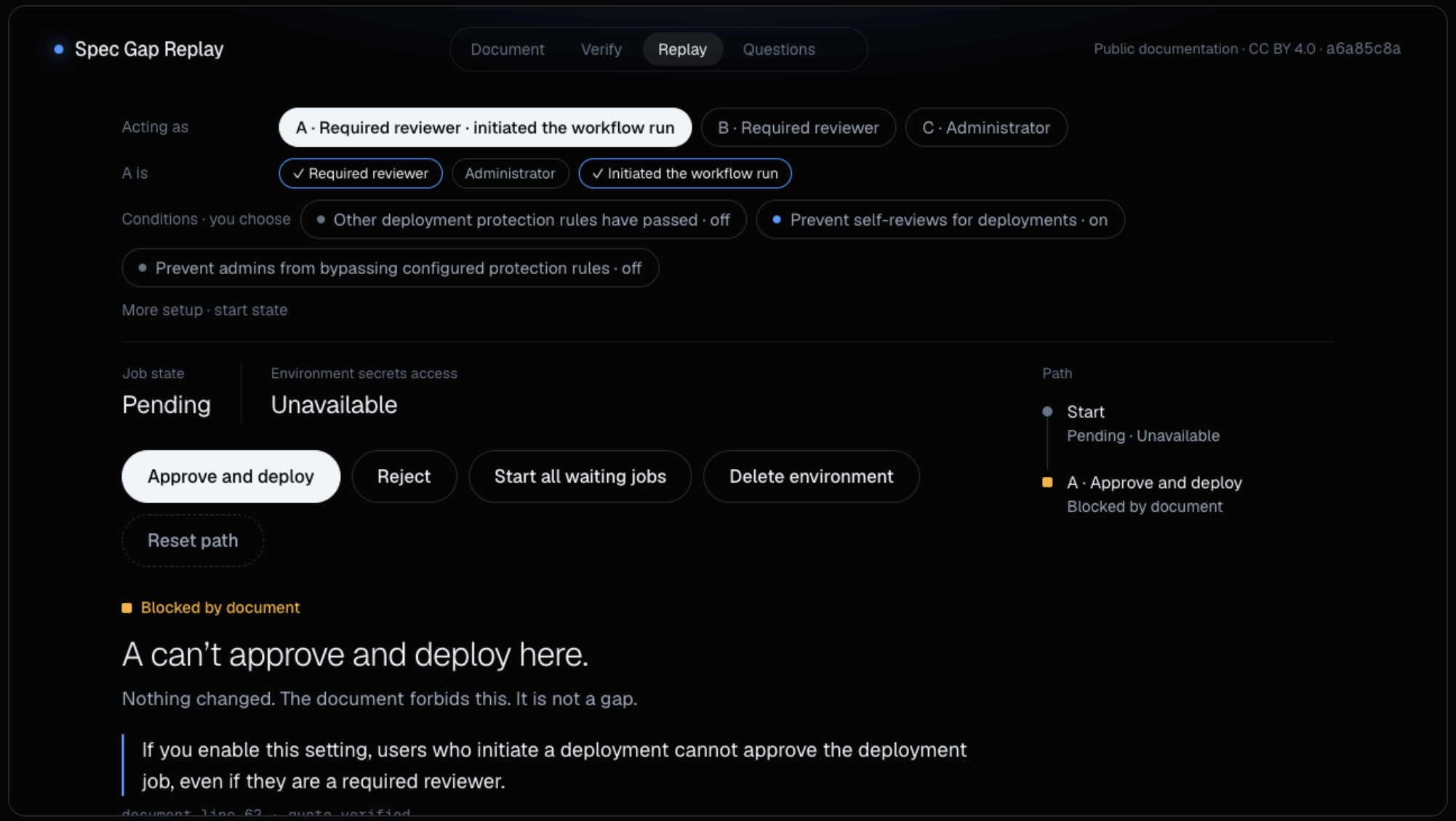
Interactions API · repairs link back with `previous_interaction_id` · deployed on Google Cloud Run

Live demo

GitHub Docs (real, CC BY 4.0) → Blocked vs Defined
Authored test spec → Needs decision → question

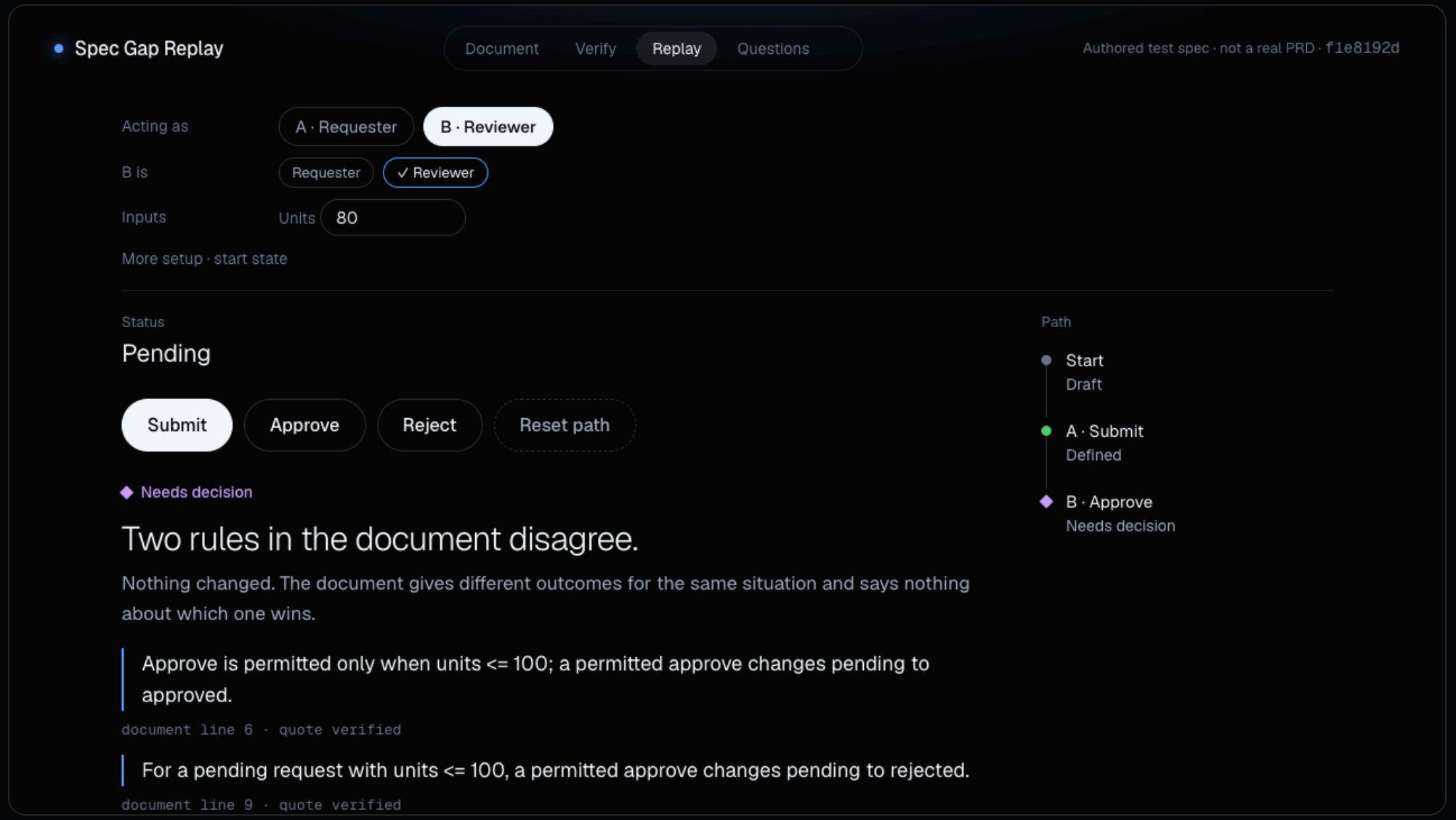
`spec-gap-replay-1088859591404.asia-northeast3.run.app`

Real GitHub Docs · same click, different setting



"Prevent self-reviews" on → Blocked by document, quoting document line 62. Turn it off → Defined.

Authored test spec · two rules disagree



Needs decision: nothing changes, both lines are cited, and gemini-3.8-flash drafts the question a person edits and exports.

Why it’s different from asking a chatbot

It doesn’t guess.

The model only extracts structure. Code applies the reviewed rules and stops when no rule decides.

Results come with their source.

Defined and Blocked results quote the rule they used. Code checks each quote exists in the document.

Forbidden is not missing.

“The document blocks this” and “the document says nothing” are different results.

Honest limits

The real GitHub Docs paths we fixed in advance showed no gaps.

The gap scene uses an authored test spec, labeled in the app.

Extraction varies between runs.

That is why a person reviews every rule before replay.

Quote checks prove the text exists, not that the meaning is right.

`github.com/currenjin/2026-google-deepmind-ai-hackathon`